

Neuro-Symbolic AI: Explainability, Challenges, and Future Trends

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Explainability is an essential reason limiting the application of neural networks in many vital fields. Although neuro-symbolic AI hopes to enhance the overall explainability by leveraging the transparency of symbolic learning, the results are less evident than imagined. This article proposes a classification for explainability by considering both model design and behavior of 191 studies from 2013, focusing on neuro-symbolic AI, hoping to inspire scholars who want to understand the explainability of neuro-symbolic AI. Precisely, we classify them into five categories by considering whether the form of bridging the representation differences is readable as their design factor, if there are representation differences between neural networks and symbolic logic learning, and whether a model decision or prediction process is understandable as their behavior factor: implicit intermediate representations and implicit prediction, partially explicit intermediate representations and partially explicit prediction, explicit intermediate representations or explicit prediction, explicit intermediate representation and explicit prediction, unified representation and explicit prediction. We also analyzed the research trends and three significant challenges: unified representations, explainability and transparency, and sufficient cooperation from neural networks and symbolic learning. Finally, we put forward suggestions for future research in three aspects: unified representations, enhancing model explainability, ethical considerations, and social impact.

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Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 INTRODUCTION

The difficulties of explaining neural networks come from its feature extraction and reasoning process based on features. It could be more intuitive to associate float numbers with our macroscopically perceived object features. Although many attempts have been made to explore this aspect, it still takes much effort to solve this problem fundamentally. Explainability is less evident than other metrics for measuring model performance, such as accuracy, recall or precision.

Facchin [69] proposed that researchers have yet to directly observe or identify a clear neural structure that fully meets or represents a specific NSR (neural structure representation) standard in neuroscience. In other words, although significant progress has been made in understanding how the brain processes and stores information, we still need to be able to specify an exact neural structure or network of neurons that precisely corresponds to a specific encoding of

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information. In addition, van der Burght et al. [183] discovered that the significant correlation between computational models and the human brain does not prove they implement the same process.

Yang et al. [200] pointed out that the current field of XAI (explainable artificial intelligence) needs a precise computational theoretical framework to explain how humans understand the explanation of AI, which limits the development of theoretical construction of XAI systems. They argue that humans expect AI to make decisions independently and understand AI's explanations by comparing them to their knowledge. This comparison process can be formalized using the universal generalization laws discovered by Shepard in similar spaces. Shepard [163] proposed a theoretical framework to predict how humans conclude AI explanations, achieving a more accurate and effective explanation method consistent with human cognitive processes. Fedyk and Ray [71] observed that many efficient machine-learning algorithms belong to the "Rashomon set." Although these models have similar performance indicators, they may differ significantly in structure or interpretation. This diversity suggests that even though some current models, as analogs of human cognitive processes, may provide a basis for generating hypotheses about human cognition, it is difficult to make accurate predictions about them because of current explainability limitations.

Broniatowski et al. [34] illustrated the different but complementary roles that explainability and understandability play in making machine learning systems more transparent and trustworthy to users. The two are not mutually exclusive but provide solutions for different aspects of user interaction with machine learning systems. Interpretability focuses on describing in detail the internal mechanisms or processes by which a model makes specific decisions to allow the model's designers or professionals to understand and verify the model's behavior. Comprehensibility emphasizes the meaning of model's output to users, that is, how to transform the output into information or a decision-making basis that is of practical significance to users. Verhagen et al. [190] proposed from the cognitive psychology perspective that special attention should be paid to the four components of perception, semantics, intention, user, and context when designing explanations. This analysis means that an additional process may need to be introduced between the original explanation generated by the AI system and the explanation ultimately provided to the user, as the former may need to be more technical and complex to understand for non-expert users. This step transforms the original explanation into a more humane and easy-to-understand form. This research inspires us to balance the technical transparency of the model with the explainability of user expectations.

The above studies show that we should be cautious about the explainability of neuro-symbolic AI, even though the logic of symbolic learning is often transparent and readable. This survey analyzes 191 neuro-symbolic AI studies from the past decade to provide research references on explainability for scholars who wish to conduct related research. Chapter two starts with an analysis of the models' explainability and then divides these studies into five categories from the perspective of design and behavior. Chapter three proposes a more detailed classification description and statistics with 14 representative cases for discussion. Then, Chapter four briefly analyzed the general trends in neuro-symbolic AI based on collected articles. Next, from Chapters Five to Six, we analyze the main research challenges and promising future research directions.

2 EXPLAINABILITY ANALYSIS

Calegari et al. [36] divided models' transparency into explainability by design and post-hoc. The former designs the models to be easily understood by humans or embeds logical rules and constraints in the model construction process to make the neuro-symbolic AI system transparent and explainable by design, essentially avoiding the impact of black box models from the beginning. The latter refers to analyzing the model's behavior to explain its decision-making process

after the system has been designed, developed, and deployed, such as by generating explanatory text, visualization technology, and model simplification.

Combining these two ideas, we proposed an explainability classification method that considers both the design and behavior of the models. As Table 1 shows, we utilize this method to classify 191 research into five categories: low, medium-low, medium, medium-high, and high.

First, we consider the difference between features extracted by neural networks and the form of information processed by symbolic logic. This perspective focuses on the compatibility and conversion mechanism between these two diagrams. In addition to the explainability of neural network itself, the format of this conversion also affects the explainability of integrated models, contributing to the first standard for categorization, which measures the readability of intermediate representation that bridges the neural representation and the logical symbolic representation. Second, we consider the explainability of decision-making or prediction logic in neuro-symbolic AI models. Specifically, even considering the inevitable impact of the black-box processing of neural networks, we still can understand the essence of knowledge-processing methods to various degrees, which will help us to explain the decision-making or prediction to a certain degree. Based on the above conditions, our specific classification method is shown in Figure 1.

Table 1. Neuro-Symbolic AI's classification by explainability

Explainability	Implicit Intermediate Representations	Partially Explicit Intermediate Representations	Explicit Intermediate Representations	Explicit Intermediate Representations	Unified Representation
Implicit Decision-Making(prediction)	and (Category I)				
Partially Explicit Decision-Making(prediction)					
Explicit Decision-Making(prediction)		and (Category II)			
Explicit Decision-Making(prediction)					
Explicit Decision-Making(prediction)			or (Category III)		
Explicit Decision-Making(prediction)					
Explicit Decision-Making(prediction)				and (Category IV)	
Explicit Decision-Making(prediction)					
Explicit Decision-Making(prediction)					and (Category V)
Explicit Decision-Making(prediction)					

3 EXPLAINABILITY-BASED CLASSIFICATION OF NEURO-SYMBOLIC METHODS

3.1 Implicit Intermediate Representations and Implicit Decision Making(prediction)

This category includes 74 neuro-symbolic AI studies with three common characteristics, as Table 2 shows. First, they all use neural networks to extract features from data. However, the representations of these features cannot be directly processed by symbolic logic, so they need an intermediate representation to fill the gap. Secondly, the intermediate representation bridging the two is often latent vector embedding or combined with structured representation but is only partially explicit and directly human-readable. Third, most of the overall decision-making logic or prediction method is implicitly expressed through the weights and activation functions of the neural network. Some methods directly integrate symbolic logic into the decision-making process or provide an indirect understanding of decision-making logic by designing interpretable interfaces such as attention mechanisms and logic rule generators. However, decision-making logic still needs to be explained overall.